Marisa Nieto Del Rio De Astis

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EDUCATION

University of Utah | Salt Lake City, UT **Expected May 2027**
B.S. in Software Development
B.S. in Environmental & Sustainability Studies (Air, Water, Health)
GPA: 3.5 **|** Dean's List | 2x Undergraduate Research Opportunities Program (UROP) Scholar

TECHNICAL SKILLS

Languages: Python (NumPy, Pandas, Matplotlib), C/C++, MATLAB, Java, SQL, C#, JavaScript, Bash
Web, Data & Cloud: React, Node.js, Django, PostgreSQL, MongoDB, Supabase, Docker, AWS, REST APIs
Embedded & Instrumentation: Arduino, Raspberry Pi, sensor integration, hyperspectral imaging, EPICS, PyDM
Tools: Linux, Git, GIS, Agile, Modbus/RS-232, pytest, CI/CD

RELEVANT EXPERIENCE

NSF REU Electrical and Computer Engineering at MSU | Bozeman, MT **May — Aug. 2026**
Undergraduate Research Assistant - National Science Foundation

- Designed and built a recirculating algae testbench (pump-driven flow tank, custom dosing replicating river and bloom chemistry from DI water), enabling in-lab hyper/multispectral imager testing without field blooms and controlled lab experiments varying depth, turbidity, and percent cover
- Built an Arduino Uno WiFi water-quality monitor (conductivity as a nutrient proxy, pH, turbidity, temperature sensors) with a self-hosted web dashboard for continuous monitoring/reports, providing ground truth for imager data
- Validated setup outdoors with Pika L hyperspectral imager: algae, diatom, and substrate spectra were consistent with field data, enabling classifier fine-tuning. Supported a peer's depth-from-spectra imagery scripts

John A. Moran Eye Center | Salt Lake City, UT **Sept. 2025 — Present**
Undergraduate Research Assistant - Data Analysis & Machine Learning

- Built a MATLAB pipeline that converts XML stimulus movies into spike-triggered-average spatial maps and fits a Linear-Nonlinear-Poisson (LNP) model to multielectrode array data from rat and tree shrew retinas
- Added a GLM stage that predicts spike trains from stimulus alone across repeated movies, extending the model from rat to tree shrew to support glaucoma research
- Wrote a Python tool automating OCT layer segmentation; built Linux/Bash pipelines for large-scale data transfer and analysis of the lab's retinal databases

SLAC National Accelerator Laboratory | Menlo Park, CA **June 2024 — Present**
Software Engineer, LCLS User Operations (Aug. 2024 — Present) · Control System Intern (June — Aug. 2024)

- Built a locally hosted prototype of a new proposal-submission portal (Django, React, PostgreSQL, Docker) that demonstrated feasibility to lab leadership and restarted a stalled cross-facility, multi-year modernization effort
- Benchmarked and refined two peer-lab proposal systems on self-hosted Linux systems, cataloged feature gaps, and gathered requirements from 4 beamline user facilities for a universal proposal system
- Automated data workflows with Google Apps Script (JS), migrating hundreds of GB of legacy proposal and conference records into a new storage system
- Converted hundreds of legacy Java/EDM screens to Python/PyDM on the existing EPICS backend; interviewed control-room operators to shape a redesign of the beamline control panel and required DBMS changes

Robotics Teaching Assistant | Salt Lake City, UT **Jan. 2023 — June 2024**
Our Lady of Lourdes Robotics Club Teaching Assistant

- Taught 20 students weekly; wrote C/C++ and MATLAB lesson code for Arduino and Raspberry Pi robot kits

PROJECTS

LiCam | *Capstone team of 4* **Jan. 2026 — Present**

- Proposed, designed, and developed a handheld depth camera pairing an IR projector/receiver (LiDAR) and RGB camera on a Raspberry Pi, streaming depth maps over BLE to a React/Supabase mobile app with 3D visualization and editing

U of U She Rocks Web Platform | *Full-Stack Developer & Club Leader* **Aug. 2025 — Present**

- Designed, built, and maintained a full-stack platform (React, Django REST, PostgreSQL, Docker) for a 150 member club, with lottery and FCFS event registration, waitlists, and officer scheduling. Wrote 200+ automated tests with CI on the production Docker image; designed the AWS production architecture